

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1706

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4
Marks:40

Total

Annexure A

CASE STUDY

Code of Conduct:

I Rekha S K (192511023) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Rekha S K

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- li. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- lii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understan ding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well- structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Answers

CASE STUDY 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Let:

- F = Patient has Fever
- C = Patient has Cough
- B = Patient has Breathlessness
- Flu = Patient has Possible Flu
- M = Patient has Possible Measles
- P = Patient has Possible Pneumonia

Rules

Rule 1:

$$F \wedge C \rightarrow Flu$$

Rule 2:

$$F \wedge Rash \rightarrow M$$

where Rash means the patient has a rash.

Rule 3:

$$C \wedge B \rightarrow P$$

Facts about Patient A

$$F = \text{True}$$

$$C = \text{True}$$

$$B = \text{True}$$

Therefore:

$$F \wedge C \wedge B$$

(b) Forward Chaining

Forward chaining starts from the known facts and applies rules to derive new facts.

Initial Facts

$$\text{Fever}(A) = \text{True}$$

$$\text{Cough}(A) = \text{True}$$

$$\text{Breathlessness}(A) = \text{True}$$

Step 1

Rule 1:

$$\text{Fever} \wedge \text{Cough} \rightarrow \text{Possible Flu}$$

We have:

Fever = True

Cough = True

Therefore:

Possible Flu = True

Step 2

Rule 3:

$\text{Cough} \wedge \text{Breathlessness} \rightarrow \text{Possible Pneumonia}$

We have:

Cough = True

Breathlessness = True

Therefore:

Possible Pneumonia = True

Rule 2

$\text{Fever} \wedge \text{Rash} \rightarrow \text{Possible Measles}$

We do not have:

Rash = True

Therefore, Measles cannot be derived.

Final Conclusions

Fever = True

Cough = True

Breathlessness = True

Possible Flu = True

Possible Pneumonia = True

(c) Backward Chaining for Pneumonia

Goal

Possible Pneumonia

Find a rule that can produce Pneumonia:

$\text{Cough} \wedge \text{Breathlessness} \rightarrow \text{Possible Pneumonia}$

So the goal is reduced into two subgoals:

Possible Pneumonia

|

↓

Cough $\wedge$ Breathlessness

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↓ ↓

Cough Breathlessness

| |

TRUE TRUE

Both subgoals are known facts.

Therefore:

Possible Pneumonia = TRUE

Conclusion

Patient A has Possible Pneumonia according to the given rule base.

CASE STUDY 2 — Autonomous Traffic Management Agent

(a) Unification

Rule 1

$\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Given facts:

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

We need to match:

$\text{Vehicle}(x)$

with:

$\text{Vehicle}(\text{Ambulance})$

Therefore:

$\theta_1 = \{x / \text{Ambulance}\}$

Now match:

$\text{EmergencyType}(x)$

with:

$\text{EmergencyType}(\text{Ambulance})$

Again:

$\theta_2 = \{x / \text{Ambulance}\}$

Final MGU

$\theta = \{x / \text{Ambulance}\}$

Therefore Rule 1 becomes:

$\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance})$

$\rightarrow \text{ClearPath}(\text{Ambulance})$

Since both facts are true:

$\text{ClearPath}(\text{Ambulance})$

is derived.

(b) Resolution Proof for Proceed(CarA)

We need to prove:

$\text{Proceed}(\text{CarA})$

Step 1: Convert Rule 1 to CNF

Original:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Remove implication:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Step 2: Rule 2

Original:

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

This gives two clauses:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

and

$\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

Step 3: Rule 3

Original:

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

CNF:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

Vehicle(CarA)

Behind(CarA, Ambulance)

To prove:

Proceed(CarA)

Negate the goal:

$\neg$ Proceed(CarA)

Resolution Step 1

From Rule 1 and:

Vehicle(Ambulance)

EmergencyType(Ambulance)

derive:

ClearPath(Ambulance)

Resolution Step 2

Using Rule 2:

$\neg$ ClearPath(x) $\vee$ GreenSignal(x)

and:

ClearPath(Ambulance)

derive:

GreenSignal(Ambulance)

Resolution Step 3

Rule 3:

$\neg$ Vehicle(x) $\vee$ $\neg$ Behind(x,y) $\vee$ $\neg$ GreenSignal(y) $\vee$ Proceed(x)

Substitute:

x = CarA

y = Ambulance

We get:

$\neg$ Vehicle(CarA) $\vee$ $\neg$ Behind(CarA,Ambulance)

$\vee$ $\neg$ GreenSignal(Ambulance) $\vee$ Proceed(CarA)

Using the facts:

Vehicle(CarA)

Behind(CarA,Ambulance)

GreenSignal(Ambulance)

we derive:

Proceed(CarA)

Step 4

We have:

Proceed(CarA)

and the negated goal:

$\neg$ Proceed(CarA)

Resolving them gives:

□

Conclusion

Proceed(CarA) is TRUE.

(c) Two Simultaneous Emergency Vehicles

The current KB can handle multiple emergency vehicles individually, because Rule 1 is universally quantified:

$\forall x$

So if we have:

Vehicle(Ambulance1)

EmergencyType(Ambulance1)

Vehicle(Ambulance2)

EmergencyType(Ambulance2)

we can derive:

ClearPath(Ambulance1)

ClearPath(Ambulance2)

However, the KB does not explicitly handle conflicts between two emergency vehicles, such as:

- Both vehicles requesting the same road.
- Which vehicle gets priority?
- Collision avoidance.
- Different emergency priorities.
- Two vehicles approaching the same intersection.

Additional Rules Needed

For example:

$\text{EmergencyVehicle}(x) \wedge \text{HigherPriority}(x,y) \rightarrow \text{GivePriority}(x)$

and:

$\text{Vehicle}(x) \wedge \text{Vehicle}(y) \wedge \text{SamePath}(x,y) \rightarrow \text{AvoidCollision}(x,y)$

Also, facts such as:

$\text{HigherPriority}(\text{Ambulance1}, \text{Ambulance2})$

SamePath(Ambulance1,Ambulance2)

would be required.

Conclusion

The current KB is sufficient for basic independent emergency routing, but additional rules are needed for conflict resolution and simultaneous emergency vehicles.

CASE STUDY 3 — Agricultural Expert Reasoning System

(a) Convert into CNF

P1

Original:

SoilDry $\rightarrow$ IrrigationNeeded

Remove implication:

$\neg$ SoilDry $\vee$ IrrigationNeeded

So CNF is:

$\neg$ SoilDry $\vee$ IrrigationNeeded

P2

Original:

IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Remove implication:

$\neg$ (IrrigationNeeded $\wedge$ CropWheat) $\vee$ ApplyDripMethod

Move negation inward:

($\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat) $\vee$ ApplyDripMethod

Therefore:

$\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod

P3

Original:

$\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Remove implication:

$\neg$ ($\neg$ ApplyDripMethod $\wedge$ CropWheat) $\vee$ CropAtRisk

Move negation inward:

ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk

Therefore:

ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk

Facts

SoilDry = True

CropWheat = True

CNF:

SoilDry

CropWheat

Final CNF Set

1. $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$
2. $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
3. $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$
4. SoilDry
5. CropWheat

(b) Resolution Refutation for ApplyDripMethod

Goal

ApplyDripMethod

Negate the goal:

$\neg \text{ApplyDripMethod}$

Add it to the KB.

Resolution Step 1

Use:

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

and:

SoilDry

Resolve:

IrrigationNeeded

Resolution Step 2

Use:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

and:

IrrigationNeeded

Resolve:

$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Resolution Step 3

Use:

$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$

and:

CropWheat

Resolve:

ApplyDripMethod

Resolution Step 4

We have:

ApplyDripMethod

and the negated goal:

$\neg \text{ApplyDripMethod}$

Resolve:

□

Resolution Proof Tree

ApplyDripMethod

|

|

|

$\neg \text{CropWheat} \vee \text{ApplyDrip}$ CropWheat

|

IrrigationNeeded

|

SoilDry + P1

Finally:

ApplyDripMethod

+

$\neg \text{ApplyDripMethod}$

↓

□

Conclusion

Since the negation of the goal leads to the empty clause:

□

the theorem is proved:

ApplyDripMethod = TRUE

(c) Remove SoilDry

Initially:

SoilDry

CropWheat

If SoilDry is removed:

CropWheat

remains.

Step 1

P1 is:

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

But we no longer have:

SoilDry

Therefore we cannot derive:

IrrigationNeeded

Step 2

P2 requires:

$\text{IrrigationNeeded} \wedge \text{CropWheat}$

We only know:

CropWheat

Therefore:

ApplyDripMethod

cannot be derived.

Step 3

P3 requires:

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat}$

But we do not have:

$\neg \text{ApplyDripMethod}$

Therefore:

CropAtRisk

also cannot be derived.

Final Conclusion

Without SoilDry:

IrrigationNeeded $\rightarrow$ Cannot derive

ApplyDripMethod $\rightarrow$ Cannot derive

CropAtRisk $\rightarrow$ Cannot derive

Important: Not being able to prove ApplyDripMethod does not mean $\neg$ ApplyDripMethod is true.

CASE STUDY 4 — Wumpus World Knowledge Agent

(a) Belief State and Modus Ponens

Given Percepts

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Rule 1

Stench[1,2] $\rightarrow$ Wumpus is adjacent to [1,2]

Since:

Stench[1,2] = TRUE

we derive:

Wumpus is adjacent to [1,2]

Rule 2

Breeze[1,1] $\rightarrow$ Pit is adjacent to [1,1]

Since:

Breeze[1,1] = TRUE

we derive:

Pit is adjacent to [1,1]

Rule 3

Glitter[x,y] $\rightarrow$ Gold[x,y]

Given:

Glitter[2,2]

therefore:

Gold[2,2]

Rule 4

$\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

We do not have enough information to prove either:

$\neg \text{Wumpus}[x,y]$

and:

$\neg \text{Pit}[x,y]$

for specific cells.

Therefore, we cannot derive Safe for any particular cell from Rule 4 alone.

Derived Knowledge

Stench[1,2]

Breeze[1,1]

Glitter[2,2]

Wumpus is adjacent to [1,2]

Pit is adjacent to [1,1]

Gold[2,2]

(b) Possible Locations of Wumpus and Pit

Assuming the usual Wumpus World grid where adjacent means up, down, left, and right:

Wumpus

Stench is at:

[1,2]

Its possible adjacent cells are:

[1,1]

[1,3]

[2,2]

Therefore:

$\text{Wumpus} \in \{[1,1], [1,3], [2,2]\}$

Pit

Breeze is at:

[1,1]

Its adjacent cells are:

[1,2]

[2,1]

Therefore:

$\text{Pit} \in \{[1,2], [2,1]\}$

Reasoning

$\text{Stench}[1,2]$

↓

Wumpus is adjacent to [1,2]

↓

Possible Wumpus locations:

[1,1], [1,3], [2,2]

And:

$\text{Breeze}[1,1]$

↓

Pit is adjacent to [1,1]

↓

Possible Pit locations:

[1,2], [2,1]

Note: The given KB does not contain enough information to determine exactly which of these cells contains the Wumpus or Pit.

(c) Is the Path Safe?

Path:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

We need to examine each cell.

Cell [1,1]

We have:

$\text{Breeze}[1,1]$

This tells us that a Pit is adjacent to [1,1].

Possible pit locations include:

[1,2], [2,1]

Therefore, [1,1] itself is not proved to contain a pit.

But we cannot prove it completely safe using Rule 4 because we don't have both:

$\neg \text{Wumpus}[1,1]$

$\neg \text{Pit}[1,1]$

Cell [1,2]

We have:

Stench[1,2]

This means:

Wumpus is adjacent to [1,2]

Possible Wumpus locations include:

[1,1], [1,3], [2,2]

Also, because of the Breeze at [1,1], [1,2] is a possible Pit location.

Therefore, [1,2] cannot be confirmed safe.

Cell [2,2]

We know:

Glitter[2,2]

Therefore:

Gold[2,2]

But the stench at [1,2] means [2,2] is also a possible Wumpus location.

Therefore, [2,2] cannot be confirmed safe from the given information.

Final Conclusion

The path:

[1,1] → [1,2] → [2,2]

is not guaranteed to be safe.

The agent knows that gold is at [2,2], but it does not have enough information to prove that [1,2] and [2,2] are free from Wumpus/Pit hazards.

Final Answer

Gold[2,2] = TRUE

Safe[1,2] = NOT PROVABLE

Safe[2,2] = NOT PROVABLE

Therefore, the path is NOT CONFIRMED SAFE.